

PC-03 · EMULSIFIERS & THICKENERS · PERSONAL CARE

HS Tween 80

Polysorbate 80

INCI	CAS	EC	FAMILY	SUPPLIED AS
Polysorbate 80	9005-65-6	—	PC-03	Per grade

01 POSITIONING

The HLB toolbox, supplied by grade — specified, verified, repeated.

02 HOW IT WORKS

01 HLB engineering

EO count sets the hydrophilic-lipophilic balance: each grade is a point on the HLB scale, combined to match the oil phase.

02 Interfacial film

Adsorbs at the oil–water boundary, dropping interfacial tension and stabilizing droplets against coalescence.

03 Electrolyte robustness

Non-ionic character keeps performance in hard water and salt-rich systems.

03 APPLICATIONS

- Creams and lotions (O/W)
- Fragrance solubilization in toners and micellar waters
- Bath oils
- Household and I&I emulsions
- Pharma-adjacent excipients (per grade)

04 FORMULATION GUIDE

USE LEVEL	0.5–6%
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PH & CONDITIONS	Stable pH 3–10
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— Match emulsifier pair HLB to the required HLB of the oil phase — the classic low+high combination beats any single grade.

— For solubilization, premix solubilizer and fragrance 3:1 to 5:1 before adding to water.

05 TYPICAL SPECIFICATION

Appearance	per grade: liquid / paste / flakes
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HLB	per grade
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Water content	≤ 1–3%
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1,4-Dioxane	≤ 10 ppm (per TDS)
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Typical parameters. The specification on the current TDS and the per-lot Certificate of Analysis govern.

06 PERFORMANCE — DIRECTIONAL

Emulsion stability (45 °C, class data)

Paired HLB system



Fragrance solubilization capacity

PEG-40 HCO systems



Values are representative of the ingredient class, based on published technical literature, and shown directionally for comparison. Final performance must be validated in the customer's formulation.